

Correlation & Covariance

Today's Big Question: When one thing changes, does another thing change too? And if they DO move together, how strong is that relationship?

Why This Matters

Correlation is one of the most important concepts in data science. Every machine learning model, every feature selection step, and every business analysis starts with correlation. You need to understand which variables are related to each other — and how strongly.

By the End of Today, You Will Be Able To:

- Understand the difference between covariance and correlation
- Calculate and interpret correlation coefficients
- Read a correlation matrix — you'll see this in every real job
- Know what strong vs. weak relationships look like
- Use correlation for feature selection in ML models
- Avoid the biggest statistics mistake: Correlation $\neq$ Causation

Real-World Use Case: Netflix wants to predict which shows you'll watch. They look at correlations: users who watched Show A often watch Show B (high correlation = good feature). If a feature has zero correlation with what you watch, it's useless — they drop it. This is feature selection, and it all starts with understanding correlation.

How to Introduce This Topic

Opening Line for the Class

Start with a simple, relatable question: "If I told you that ice cream sales and shark attacks both go up in summer, does that mean ice cream causes shark attacks?" Let the room react — this is the hook that motivates the entire day.

Why This Topic Comes Before Everything Else

Almost every later topic in this course — regression, hypothesis testing, machine learning features — depends on knowing how two variables relate. Correlation and covariance are the foundation. If a student doesn't understand this, they will struggle with regression next week, because regression literally builds a line through correlated data.

Two Ideas We're Separating Today

- **Direction** — do the variables move the same way or opposite ways? (this is what covariance answers)
- **Strength** — how tightly do they move together? (this is what correlation answers)
- Most people confuse these two ideas. By the end of today, your students should be able to explain the difference in one sentence each.

A Simple Analogy to Use Out Loud

Think of two people dancing. **Covariance** tells you if they're moving in sync or out of sync (same direction vs. opposite direction). **Correlation** tells you how well-rehearsed the dance is — are they moving in perfect lockstep, or just loosely following each other? Covariance can't tell you that because its scale changes depending on how big the dance floor is (the units). Correlation fixes the scale so you can compare any two dancers fairly.

PART 1

Covariance — Do They Move Together?

Definition: Covariance measures if two variables move in the **same** direction or **opposite** directions. It tells you the direction of a relationship, but not how strong it is.

Think of it as the very first, roughest check you can do on two variables: do they generally rise and fall together, or does one rise while the other falls? Covariance does not tell you how reliable or tight that pattern is — it only tells you which way the wind is blowing.

Three Possible Scenarios

Scenario	What Happens	Covariance Value	Real Example
Positive Covariance	Both go up together (or both go down together)	Positive (> 0)	Temperature up → Ice cream sales up
Negative Covariance	One goes up, the other goes down	Negative (< 0)	Temperature down → Umbrella sales up
Zero Covariance	No pattern; they move independently	0 (or close to it)	Age vs. shoe size (independent)

Talking Point: Why 'Direction Only' Matters

Emphasize to the class that covariance answers a yes/no/sideways question, not a 'how much' question. A covariance of 5 and a covariance of 5,000 could represent the exact same real-world relationship — the number by itself is meaningless until you know the units and scale involved. This is the exact problem that correlation will solve later in the session.

Covariance — Real Business Examples

Example 1: Marketing Spend vs. Revenue (Positive Covariance)

When you spend more on marketing, revenue tends to go up.

Month	Ad Spend	Revenue
Month 1	Rs 10K	Rs 100K
Month 2	Rs 15K	Rs 120K
Month 3	Rs 20K	Rs 140K

Both variables increase together → **Positive covariance**. Talking point: notice spend roughly doubles (10K→20K) while revenue only grows by 40% (100K→140K). Covariance won't tell you that the relationship is less than proportional — for that you'd need correlation and, later, regression. For now, just flag that both move the same way.

Example 2: Price vs. Quantity Sold (Negative Covariance)

When you increase the price, quantity sold tends to go down.

Week	Price	Units Sold
Week 1	Rs 100	500
Week 2	Rs 150	300
Week 3	Rs 200	100

One increases while the other decreases → **Negative covariance**. Talking point: this is the classic demand curve every economics student learns — price up, demand down. Ask the class: what other pairs of variables in business naturally move in opposite directions? (e.g., discount % vs. profit margin, commute distance vs. property price).

Example 3: Hours Worked vs. Customer Complaints (Zero Covariance)

Worker A: 8 hrs → 2 complaints | Worker B: 6 hrs → 3 complaints | Worker C: 10 hrs → 2 complaints

No clear pattern → **Zero covariance**. Talking point: this is an important real-world lesson — just because two things *could* be related (you might assume tired workers make more mistakes) doesn't mean the data shows it. Use this to set up tomorrow's reminder: always check the data, don't just trust intuition.

PART 1 (CONTINUED)

Covariance — The Math & The Problem

The Math (Don't Panic!)

Covariance = Average of [(each X value – mean of X) × (each Y value – mean of Y)]

In plain English: for each observation, calculate how far X is from its mean, how far Y is from its mean, multiply them together, then average all those products.

Worked Example (Walk Through This on the Board)

Using the marketing data: spend = [10, 15, 20, 12, 18], mean spend = 15. Revenue = [100, 120, 140, 110, 130], mean revenue = 120.

Month	Spend – Mean	Revenue – Mean	Product
1	10 – 15 = –5	100 – 120 = –20	100
2	15 – 15 = 0	120 – 120 = 0	0
3	20 – 15 = 5	140 – 120 = 20	100
4	12 – 15 = –3	110 – 120 = –10	30
5	18 – 15 = 3	130 – 120 = 10	30

Average of the products = (100 + 0 + 100 + 30 + 30) / 5 = **52** (population formula; numpy's default uses a slightly different divisor, hence the 38.5 you'll see in code — mention this difference if a sharp student asks). The key teaching point: every row where spend and revenue moved the same way relative to their averages contributed a *positive* product, pulling covariance higher.

Python: Calculate Covariance

```
import numpy as np
import pandas as pd

marketing_spend = [10, 15, 20, 12, 18] # Rs thousands
revenue = [100, 120, 140, 110, 130]

covariance = np.cov(marketing_spend, revenue)[0, 1]
print(f"Covariance: {covariance}") # Output: 38.5 (positive)
```

```
# Using pandas (easier!)
df = pd.DataFrame({'spend': marketing_spend, 'revenue': revenue})
cov_matrix = df.cov()
print(cov_matrix)
```

The Biggest Problem with Covariance

The size of covariance depends on the **units** of your data. Covariance between age (years, 20–80) and salary (rupees, 200K–2M) will be huge numbers; the same relationship measured in thousands gives smaller numbers. This makes it impossible to compare across datasets — which is why we use **correlation** instead.

Talking point for the class: ask them, "Is a covariance of 5,000 between age and salary a strong relationship?" The honest answer is *you cannot tell* — it depends entirely on whether salary is measured in rupees or in lakhs. This single limitation is the entire motivation for the next section. Pause here and let that sink in before moving to correlation.

PART 2

Correlation — How Strong Is the Relationship?

Definition: Correlation takes covariance and standardizes it to a fixed scale: -1 to $+1$. Now you can actually compare relationships — it tells you both direction *and* strength.

Mechanically, correlation is just covariance divided by the standard deviations of both variables. That division is what cancels out the units — whether salary was in rupees or lakhs no longer matters once you divide by its own spread. This is why correlation is sometimes called 'standardized covariance.' You don't need to derive this for the class, but it's a good one-liner if someone asks *why* correlation is bounded between -1 and $+1$.

The Correlation Scale (Most Important Thing Today!)

Correlation (r)	Strength	Direction
+1.0	Perfect	Positive (almost never happens)
+0.7 to +0.9	Strong	Positive
+0.5 to +0.7	Moderate	Positive
+0.3 to +0.5	Weak	Positive
0 to +0.3	Very weak / none	Positive
0	None	No relationship at all
-0.3 to 0	Very weak / none	Negative
-0.5 to -0.3	Weak	Negative
-0.7 to -0.5	Moderate	Negative
-0.9 to -0.7	Strong	Negative
-1.0	Perfect	Negative (almost never happens)

Talking point: have the class read the table both ways — up from 0 (strength only) and noticing the sign separately (direction only). A common student error is treating -0.85 as 'worse' than $+0.40$ just because it's negative. Stress that strength is about the *distance from zero*, not whether the number is positive or negative.

Correlation — Interpretation Guide

Simple Interpretation Guide

- $r = 1$ or -1 : Perfect relationship (doesn't happen in real data)
- $r = \pm 0.7$ to ± 0.9 : **STRONG** relationship — tightly linked
- $r = \pm 0.5$ to ± 0.7 : **MODERATE** relationship — move together but not perfectly
- $r = \pm 0.3$ to ± 0.5 : **WEAK** relationship — some connection, but loose
- $r = 0$ to ± 0.3 : **VERY WEAK / NONE** — essentially independent

Note for the class: these bands are general guidelines, not laws of nature. A correlation that's 'moderate' in social science (where human behavior is noisy) might be considered weak in physics or engineering (where measurements are precise). Always ask: how much noise do we expect in this domain?

Real-World Correlation Examples

Two Variables	Likely r	Why?
Study Hours vs. Exam Score	+0.85	Strong positive: more study, better grades
Age vs. Salary (early career)	+0.60	Moderate positive: more experience over time
Exercise Hours vs. Body Weight	-0.75	Strong negative: more exercise, less weight
Temperature vs. Hot Drink Sales	-0.80	Strong negative: hot days, fewer coffee sales
Temperature vs. Ice Cream Sales	+0.85	Strong positive: hot days, more ice cream
Shoe Size vs. Intelligence	+0.05	Nearly zero: unrelated
Salespeople vs. Rainfall	~0	Zero: totally independent variables

Use this table as a quiz: cover the 'Likely r ' column and ask students to guess strong/moderate/weak before revealing the number. This builds intuition faster than just reading values off a page.

Correlation — Python & Heatmaps

Python: Calculate Correlation

```
import numpy as np
import pandas as pd

marketing = [10, 15, 20, 12, 18]
revenue = [100, 120, 140, 110, 130]

correlation = np.corrcoef(marketing, revenue)[0, 1]
print(f"Correlation: {correlation:.2f}") # Output: 0.99 (very strong positive!)

df = pd.DataFrame({
    'marketing': marketing, 'revenue': revenue,
    'ad_clicks': [50, 75, 95, 60, 85]
})
correlation_matrix = df.corr()
print(correlation_matrix)
# marketing revenue ad_clicks
# marketing 1.00 0.99 0.98
# revenue 0.99 1.00 0.97
# ad_clicks 0.98 0.97 1.00
```

Visualizing Correlations: Heatmap

When you have many columns, create a correlation heatmap. Dark red = strong positive, dark blue = strong negative, white = zero correlation.

```
import seaborn as sns
import matplotlib.pyplot as plt

sns.heatmap(correlation_matrix, annot=True, cmap='coolwarm', center=0)
plt.show()
```

Correlation $\neq$ Causation

This is the #1 mistake people make with correlation. A high correlation feels like proof, especially when the number is something dramatic like 0.90 or 0.92. But correlation only ever measures co-movement — it has no concept of cause and effect. Two variables can move together for at least three reasons: (1) one truly causes the other, (2) a third, hidden variable causes both of them, or (3) it's pure coincidence in a small sample. Correlation cannot distinguish between these three cases on its own.

Example	Correlation	The Hidden Cause (Confounding Variable)
Ice Cream Sales vs. Drowning Deaths	$r \approx +0.90$	Summer heat causes both — people buy ice cream AND swim more
Number of Firefighters vs. Fire Damage	$r = +0.92$	Bigger fires need more firefighters AND cause more damage
Shoe Size vs. Reading Ability (kids)	$r = +0.85$	Age — older kids have bigger feet AND read better

Teaching tip: for each row, ask the class to name the confounding variable themselves before you reveal it. This turns a passive slide into an active exercise and tends to be the moment students remember most from the whole session.

The Golden Rule: Correlation tells you *if* two things move together. It does not tell you *why*, or whether one causes the other.

So When Can You Say Causation?

Only in controlled experiments where you:

- Control for all confounding variables
- Randomly assign people to treatment vs. control groups
- Change only one variable and measure the effect
- Repeat the experiment multiple times
- Get peer review and statistical significance ($p < 0.05$)

Correlation in Data Analytics

Why Does Correlation Matter?

- **Feature Selection:** If a variable has zero correlation with your target, it's useless for prediction — drop it
- **Multicollinearity:** If two features are highly correlated, you might not need both; this can cause problems in ML models
- **Understanding Your Data:** Correlation helps you explore relationships before building models (EDA)
- **Business Insights:** 'Marketing spend correlates 0.85 with revenue' signals good ROI (but doesn't prove causation)
- **Risk Management:** Portfolio managers check correlations between investments to avoid duplicate risk

Where Do You Use Correlation in Your Job?

Data Analytics Task	How You Use Correlation
Exploratory Data Analysis (EDA)	Create correlation heatmaps to find patterns
Feature Selection for ML	Keep high-corr features with target, drop zero-corr features
Detecting Multicollinearity	Remove redundant features that are too similar
Business Reporting	Tell stakeholders: 'Revenue correlates 0.72 with marketing spend'
Risk Assessment	Check if two products/investments move together (diversification)
Anomaly Detection	If variables usually correlate but suddenly don't, something's wrong

AVOID THESE

Common Mistakes Students Make

Mistake 1: 'High correlation means causation.'

Reality: No. You need controlled experiments to prove causation. Correlation just shows association.

Mistake 2: ' $r = 0.45$ is weak, so I'll ignore it.'

Reality: Context matters! In psychology, $r = 0.45$ is decent. In physics, it's weak. Even weak correlations can matter in large datasets.

Mistake 3: 'I have 500 variables. I'll keep all of them!'

Reality: Use correlation to drop variables with zero/near-zero correlation to your target. Fewer features = simpler, faster models.

Mistake 4: 'Correlation of 0.82 and 0.83 are basically the same.'

Reality: The difference matters: $0.83^2 = 69\%$ explained, $0.82^2 = 67\%$ explained. That 2% can be significant.

Mistake 5: 'Negative correlation is bad.'

Reality: Negative just means opposite direction. $r = -0.85$ is still a strong relationship. In portfolios, negative correlations are GOOD (diversification)!

PRACTICE

Problems — Correlation & Covariance

Problem 1: You calculate correlation between 'hours slept' and 'test score' and get $r = +0.78$. What does this mean? Is this a strong relationship?

Hint: $r = 0.78$ falls in the $+0.7$ to $+0.9$ band — a strong positive relationship. More sleep tends to go with higher scores, but remind students this still isn't proof that sleep *causes* better scores.

Problem 2: Correlation between 'ice cream sales' and 'swimming pool visits' is $r = +0.89$. Does eating ice cream cause swimming? Explain the confounding variable.

Hint: the confounding variable is temperature/season — hot weather drives both ice cream sales and pool visits independently.

Problem 3: You have 5 features for your ML model. Their correlations with the target are: $+0.15$, $+0.72$, $+0.08$, -0.65 , $+0.92$. Which feature should you drop? Why?

Hint: drop the $+0.08$ feature (and consider $+0.15$) — both are very weak/near-zero correlation with the target and add little predictive value.

Problem 4: Two investment funds have correlation $r = +0.95$. Should you buy both for your portfolio? Explain.

Hint: probably not for diversification — they move almost identically, so holding both doesn't reduce risk much. You'd want a fund with low or negative correlation instead.

Problem 5: Create a correlation matrix in pandas for: age, income, years_experience, and education_years. Write the code.

Hint: `df[['age','income','years_experience','education_years']].corr()` — have students predict which pairs will be most correlated before running the code.

BONUS

Pearson vs. Spearman Correlation

Type	When to Use	Example
Pearson (r)	Standard correlation — both variables continuous and roughly normally distributed	Height vs. Weight
Spearman (ρ)	Rank correlation — data is ordinal (rankings) or skewed/not normal	Movie ranking vs. IMDb rating

```
# Pearson correlation (default)
correlation_pearson = df['x'].corr(df['y'])

# Spearman correlation
correlation_spearman = df['x'].corr(df['y'], method='spearman')

print(f"Pearson r: {correlation_pearson:.2f}")
print(f"Spearman rho: {correlation_spearman:.2f}")
# Sometimes they give slightly different results!
```


Cheat Sheet — Wednesday

Concept	Definition	Range	Key Point
Covariance	Do variables move together?	Any number	Direction only, hard to interpret
Correlation	How strong is the relationship?	-1 to +1	Standard scale, easy to compare
$r = \pm 0.7$ to ± 0.9	Strong relationship	-1 to +1	Variables move tightly together
$r = \pm 0.5$ to ± 0.7	Moderate relationship	-1 to +1	Some connection but not tight
$r = \pm 0.3$ to ± 0.5	Weak relationship	-1 to +1	Loose connection, limited prediction use
$r = 0$ to ± 0.3	Very weak or none	-1 to +1	Independent, drop from ML models
Causation	One causes the other	N/A	Requires controlled experiments only!

Key Takeaways & Next Steps

Correlation Strength Summary (Memorize This!)

Correlation Range	Strength	In Human Terms
± 0.9 to ± 1.0	Perfect / Very Strong	These are basically the same thing
± 0.7 to ± 0.9	Strong	Clear relationship, definitely connected
± 0.5 to ± 0.7	Moderate	Some relationship, but with lots of noise
± 0.3 to ± 0.5	Weak	Relationship exists but is loose
0 to ± 0.3	Very Weak / None	Basically independent, maybe ignore it
0	None	Completely unrelated

Key Takeaways from Wednesday

- Covariance shows direction; correlation shows strength
- Correlation is always -1 to $+1$ — memorize what each range means!
- Correlation $\neq$ Causation. This is the most important lesson!
- Use correlation for feature selection: drop variables with zero correlation to your target
- Create correlation heatmaps as part of your EDA (Exploratory Data Analysis)
- In real jobs, you'll see correlation matrices every day — get comfortable reading them!

Next Steps: Tomorrow (Thursday), we'll learn Hypothesis Testing and p-values — how to decide if your findings are real or just luck. This is the foundation for A/B testing and experiment design!